

The primary goal of this project is to bring powerful data mining and analytics methods, as well as computing technology and technical software, to the cardiovascular disease (CVD) research community to enable everyone, regardless of expertise, to model multimodal data for the purpose of disease prediction with the end goal of clinical decision support. Our working hypothesis is that automated machine learning (AutoML) will accelerate the development of innovative strategies for translation of research findings to clinical use by enabling everyone to analyze biomedical data using data mining methods. This project builds on our user-friendly and open-source Aliro platform that represents one of the very first AutoML methods with a memory of every model and an artificial intelligence (AI) algorithm that can learn from that experience to suggest and automatically launch new analyses. A major benefit of this approach is that it makes machine learning accessible to novice users because it takes the guesswork and complexity out of picking, running, tuning, and optimizing machine learning algorithms. Bringing this technology to the clinical and translational research communities will open the door to broad adoption of data mining methods for embracing the complexity of the relationship between multimodal biomarkers and clinical outcomes. We propose here novel algorithms to adapt and extend Aliro for the large volumes of multi-omics data that are being collected by the NHLBI-funded Trans-Omics for Precision Medicine (TOPMed) program. Specifically, we will first develop an ontology-based Cardiovascular KnowledgeBase (CardioKB) tailored to multi-omics data and their relationships with cardiovascular diseases to inform the machine learning algorithms and assist with interpretation (AIM 1). We will then develop a Cardiovascular Graph (CardioGraph) algorithm to prioritize the selection of multi-omics biomarkers for machine learning analysis using measures of network topology applied to the knowledge graph from CardioKB (AIM 2). We will then develop a Cardiovascular Artificial Intelligence (CardioAI) algorithm and open-source software for automated machine learning (AutoML) analysis of multimodal biomarker data and clinical outcomes informed by CardioKB and CardioGraph (AIM 3). We will apply CardioKB, CardioGraph, and CardioAI to building predictive models of clinical outcomes using multimodal biomarkers including genetic, epigenomic, transcriptomic, metabolomic, and proteomic measures (AIM 4). Finally, we will distribute and support CardioKB, CardioGraph, and CardioAI as open-source software (AIM 5).

AIM 1. Develop a Cardiovascular KnowledgeBase (CardioKB) tailored to multi-omics data and their relationships with cardiovascular diseases. We will develop a cardiovascular-specific knowledgebase to be used by Aliro to inform the selection of multimodal biomarkers and machine learning models. We will utilize Neo4j as a graph database to manage, store, retrieve, and make probabilistic inferences about the relationships between multimodal biomarkers and biochemical pathways, genes, drugs, and health outcomes.

AIM 2. Develop a Cardiovascular Graph (CardioGraph) algorithm to prioritize the selection of multi-omics biomarkers for machine learning analysis using measures of network topology applied to the knowledge graph from CardioKB. We will develop a CardioGraph algorithm which uses measures of network topology to prioritize and extract multi-omics biomarkers from the CardioKB knowledge graph. This will provide cardiovascular-specific expert knowledge to CardioAI.

AIM 3. Develop a Cardiovascular Artificial Intelligence (CardioAI) algorithm and open-source software for automated machine learning (AutoML) analysis of multi-omics biomarker and cardiovascular outcomes data informed by the CardioKB knowledgebase. CardioAI will make use of the user-friendly and open-source Aliro software which we have previously developed and deployed. We will adapt and extend Aliro to inform multi-omics biomarker selection and model building using cardiovascular-specific knowledge from CardioKB and CardioGraph. This will make CardioAI more computationally efficient and improve the cardiovascular relevance and interpretability of the machine learning models for predicting health outcomes.

AIM 4. Apply CardioKB, CardioGraph, and CardioAI to building predictive models of health outcomes using multi-omics biomarkers including genetic, epigenomic, transcriptomic, metabolomic, and proteomic measures. All data will be access from the NHLBI-funded Trans-Omics for Precision Medicine (TOPMed) program. All machine learning models will be developed using cross-validation methods and all results replicated using holdout data the models haven't previously seen. All models will be interpreted using CardioKB and with computational methods such as Shapley values and feature importance scores.

AIM 5. Distribute and support CardioKB, CardioGraph, and CardioAI as open-source software via Github. We will distribute all software developed by this project as open source to benefit the CVD and precision health research communities.